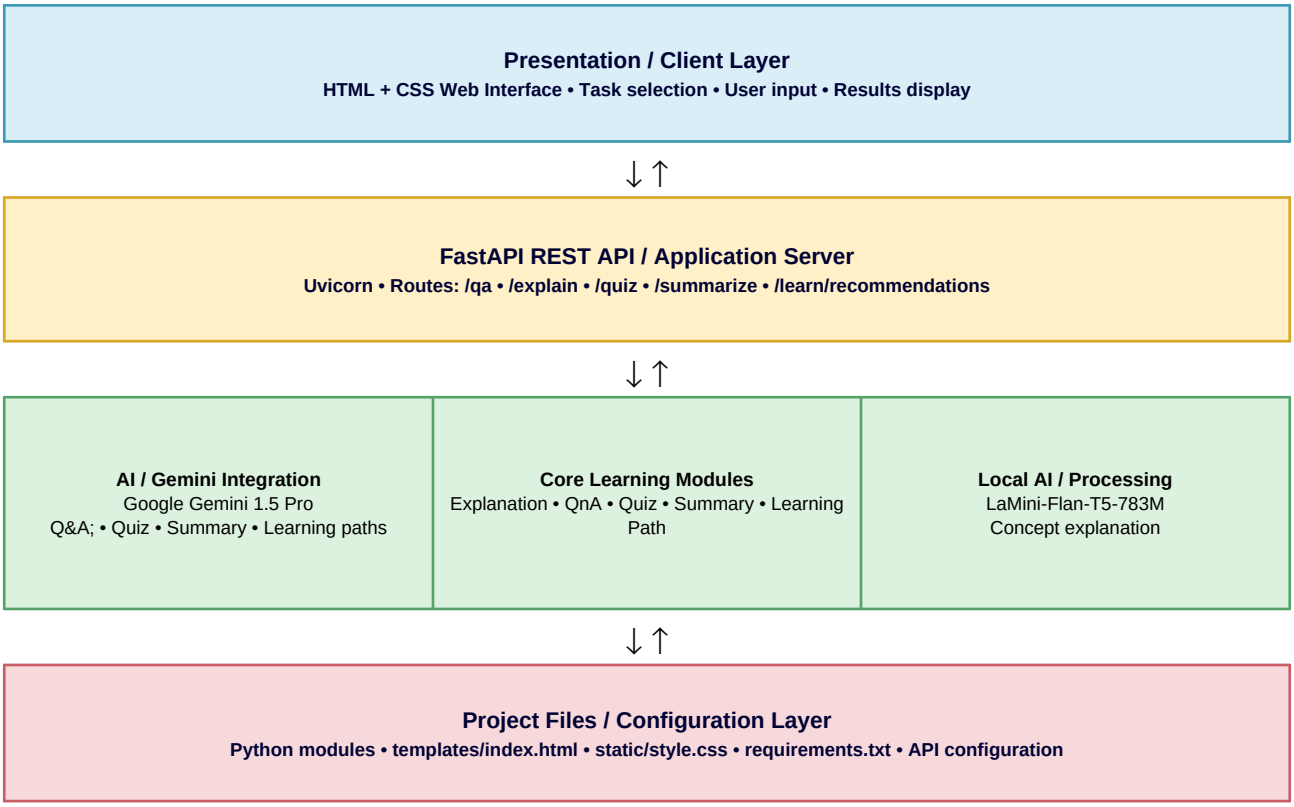
Solution Architecture

Team Enrollment Confirmation – EduGenie: Google Gemini Powered Learning Assistant

Date	01 October 2026
Team ID	SWTID-2026-1940
Project Name	Team Enrollment Confirmation - EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture for EduGenie based on the supplied project document. The system uses a simple HTML/CSS client, a FastAPI backend, Google Gemini for cloud AI tasks, LaMini-Flan-T5-783M for local concept explanation, and modular Python components.



Architecture Flow

1. User Selects a learning task and enters content/question.	**2. FastAPI** Receives the request and routes it to the correct module.	**3. AI / Modules** Gemini or local model processes the request.	**4. Result** Response is returned to the web interface.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides the web interface where users select Explain, QnA, Quiz, Summary, or Recommend Path, enter text, submit requests, and view results.	HTML, CSS
FastAPI REST API	Acts as the backend application server. It exposes RESTful endpoints and connects each endpoint to the corresponding module logic.	Python 3.10+, FastAPI, Uvicorn
Core Learning Modules	Implements concept explanation, question answering, quiz generation, summarization, and personalized learning recommendations.	Python, Gemini API, LaMini-Flan-T5-783M
Google Gemini Integration	Provides cloud-based generative AI for Q&A;, summarization, quiz generation, and learning-path recommendations.	Google Gemini 1.5 Pro API
Local Explanation Module	Uses a lightweight local generative model to explain complex topics in concise and understandable language.	LaMini-Flan-T5-783M
Project Files / Configuration	Organizes application code, frontend templates, styling, dependencies, and configuration used to run the application.	Python modules, Jinja2, HTML/CSS, requirements.txt

Team Members & Work Assignment

No.	Team Member	Assigned Work
1	**R. Priya Dharshini**	**Solution Architecture & Backend Lead:** Prepare the architecture diagram, FastAPI application structure, REST endpoints, and integration flow.
2	**S. Shobana**	**Gemini AI Integration:** Handle Google Gemini API integration for Q&A;, summarization, quiz generation, and learning recommendations.
3	**S. Rakshana**	**Core Learning Modules:** Develop and organize explanation, QnA, quiz, summary, and learning-path module logic.
4	**E. Shylaja**	**Frontend Development:** Build and style the HTML/CSS interface, task dropdown, input area, submit button, and result display.
5	**S. Sheeba**	**Testing, Deployment & Documentation:** Perform functional testing, run/deploy the application locally with Uvicorn, verify workflows, and maintain project documentation.

Key Project Workflow

• **Model Selection:** Gemini 1.5 Pro is used for Q&A;, summarization, quiz generation, and learning paths; LaMini-Flan-T5-783M is used for concept explanation.

• **Backend:** FastAPI exposes /qa, /explain, /quiz, /summarize, and /learn/recommendations endpoints.

• **Frontend:** HTML/CSS provides task selection, user input, submission, and real-time result display.

• **Testing:** Validate question answering, explanations, quiz generation, summaries, and personalized learning plans.

• **Run Locally:** Uvicorn runs the application with `uvicorn main:app --reload`.

Source basis: Prepared from the uploaded EduGenie project document, including its stated architecture, technology stack, modules, endpoints, frontend integration, and deployment workflow.